G16: Manifold-Aware Latent Diffusion Models

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Abstract

Diffusion models have been widely applied to generative tasks as they are capable of learning complex data distributions with a *score function*. In many data science problems, it may be necessary to respect the unique geometrical connectivity of the data. Moreover, accounting for the inherent geometric constraints can accelerate the inference time complexity of diffusion models. We study the learning of the geometry of data in the context of latent diffusion models. We propose a novel approach to learn a latent Symmetric Positive Definite (SPD) Riemannian metric *directly* from the data, and align it with the decoder's pull-back geometry to promote isometry. Combined with score-based diffusion on the learned latent manifold, we can generate new samples in a manifold-aware manner. The capability of learning a true metric and the latent manifold was validated with geodesic computation and data reconstruction tasks on several geometrically structured datasets.

1 Introduction

Generative modeling aims to capture complex, high-dimensional data distributions for applications ranging from image synthesis to molecular design. A particularly powerful approach, *diffusion models*, has emerged as a state-of-the-art method for learning these distributions. These models excel at capturing intricate data structures by progressively transforming noise into realistic samples through a sequence of learned denoising steps. However, most diffusion models operate in Euclidean spaces, inherently assuming that the data lie in a flat, unstructured ambient space. This assumption can significantly distort the intrinsic geometry of the data, leading to inefficient sampling and potentially degraded model performance.

Real-world data, however, often exhibit lower-dimensional, curved structures, as suggested by the well-known *manifold hypothesis*. For example, natural images, speech signals, and biological data typically reside on structured manifolds embedded in high-dimensional spaces. Failing to respect these geometric constraints can result in unnatural samples and inefficient learning. While recent work has extended diffusion models to non-Euclidean settings, these approaches often rely on predefined manifolds with fixed geometries, limiting their flexibility and generalization.

In this work, we propose a novel, fully *manifold-aware latent diffusion model* that directly learns the intrinsic geometry of the data manifold. Our approach incorporates a learnable *Symmetric Positive Definite (SPD)* Riemannian metric, trained to align the latent manifold's geometry with the decoder's pull-back metric, promoting local isometry. By coupling this learned metric with a latent diffusion process, our method can generate realistic samples that respect the underlying manifold structure without requiring strong assumptions about the data geometry.

2 Motivation

Standard latent diffusion models often operate in Euclidean latent space with no geometric constraints enforced, which can distort or completely ignore the intrinsic geometry of real-world data. This motivates us to ask the following two important questions: 1.) Can we utilize/preserve the *intrinsic geometry* of the data in the latent manifold? 2.) Can we make *minimal assumptions* about the latent manifold?

3 Related Works

Isometric Representation Learning. Representation learning embeds high-dimensional data into lower-dimensional spaces while aiming to preserve geometric and semantic structures [Tenenbaum et al., 2000]. Traditional methods like PCA and Autoencoders (AEs) often distort intrinsic manifolds due to the lack of geometric constraints [Gropp et al., 2020, Lee et al., 2022]. To address this, recent work imposes isometry via regularization. Gropp et al. [2020] proposed Isometric Autoencoders (ISO-AE), adding isometric constraints to the encoder-decoder, though their Monte Carlo-based method struggles with scalability. Lee et al. [2022] introduced more efficient, coordinate-invariant regularizations, yet exact isometry enforcement remains challenging and sensitive to tuning. However, the approach Lee et al. [2022] implicitly assumed that the Riemannian metric of the latent manifold is flat.

Riemannian Diffusion Models. Diffusion models excel in generative tasks [Vahdat et al., 2021, Yang et al., 2023] but are typically confined to Euclidean spaces. Extensions to Riemannian manifolds [Huang et al., 2022, De Bortoli et al., 2022, Zeng et al., 2023, Liu et al., 2023] improve performance on structured data by incorporating manifold-aware SDEs and sampling. Huang et al. [2022], De Bortoli et al. [2022] generalize diffusion to curved spaces, while Zeng et al. [2023] adapt it to homogeneous spaces via geometric Euler-Maruyama integration. However, these methods depend on pre-defined manifold structures, and standard latent diffusion models [Vahdat et al., 2021, Lee et al., 2023] lack geometric interpretability, limiting alignment and generalization.

4 Methodology

We propose a novel *metric parameterization* method using kernel-weighted low-rank aggregation. This approach explicitly pursues isometry and applies curvature regularization to promote the smoothness and numerical stability of the learned metric. Below, we present the problem formulation and the design of our method.

Notably, our proposed method contains a three-stage training strategy:

1. We first train a VAE with parameters (ϕ, θ) to obtain a good quality latent representation.
2. Then we freeze the pair and start the training of the metric network ψ to learn the regularized geometric structure of the VAE's latent representations.
3. Lastly, we freeze the triplet (ϕ, θ, ψ), and train a score network φ that performs latent diffusion on the learned geometric structure.

The rest of the section is organized as follows: we first formulate our problem in Section 4.1, and then we explain our proposed metric regularized learning in Section 4.2, as well as the manifold-aware latent diffusion in Section 4.3. Finally, we briefly remark on how we can compute geodesics based on the learned regularized metric in Section 4.4 for later illustrations.

4.1 Problem Formulation

Notations. Throughout the paper, we let $\mathcal{M}$ be a Riemannian manifold of dimension m with local coordinates $z \in \mathbb{R}^m$ and Riemannian metric $G(z) \in \mathbb{R}^{m\times m}$. Similarly, let $\mathcal{N}$ be a Riemannian manifold of dimension d with local coordinates $x \in \mathbb{R}^d$ and Riemannian metric $H(x) \in \mathbb{R}^{d\times d}$. Consider a smooth mapping $\mathrm{f}\colon \mathcal{M} \to \mathcal{N}$, represented in local coordinates as $f\colon \mathbb{R}^m \to \mathbb{R}^d$.[1] The differential of this mapping is the Jacobian matrix $J_f(z) := \frac{\partial f}{\partial z}(z) \in \mathbb{R}^{d\times m}$, known as the *push-forward operator*, which maps tangent vectors from $T_z\mathcal{M}$ to $T_{f(z)}\mathcal{N}$.

[1] We use f for the global, chart-independent mapping, and f for its local, chart-dependent representation.

In the context of latent diffusion models, our goal is to generate realistic samples from the unknown data distribution p_{data} of a dataset $\{x_i\}_{i=1}^n$ that resides on a smooth Riemannian manifold $\mathcal{N}$ of ambient dimension d. In the classical setting, this ambient space is typically treated as a flat Euclidean space $\mathbb{R}^d$ with a trivial metric, $H(x) = I$, which ignores the underlying geometric structure of the data.

However, according to the widely accepted *manifold hypothesis* [Cayton et al., 2008], real-world high-dimensional data often lie on a much lower-dimensional, curved manifold, which we denote by $\mathcal{M}$ with intrinsic dimension $m \ll d$. This motivates the use of a *Variational Autoencoder (VAE)* to learn a latent representation of this manifold: let $g_\phi\colon \mathbb{R}^d \to \mathbb{R}^m$ be the encoder and $f_\theta\colon \mathbb{R}^m \to \mathbb{R}^d$ be the decoder of a pretrained VAE with parameters (ϕ, θ), which together define the map

$$f_\theta \circ g_\phi\colon \mathbb{R}^d \to \mathbb{R}^d, \quad x \overset{g_\phi}{\mapsto} z \overset{f_\theta}{\mapsto} \hat{x}, \tag{1}$$

where $z = g_\phi(x)$ represents a point on the latent manifold $\mathcal{M}$. Through this mapping, f_θ serves as the smooth embedding from the latent manifold to the ambient space.

4.2 Metric Parameterization

To preserve the intrinsic geometric structure of $\mathcal{M}$ during this transformation, we seek to enforce *isometry*, a critical property ensuring that local distances and angles are preserved. A map $f\colon \mathcal{M} \to \mathcal{N}$ is an (local) isometry if and only if the pullback of the ambient metric $H(f(z))$ through f matches the intrinsic metric $G(z)$ of the latent manifold. This relationship is expressed as:

$$G(z) = J_f(z)^\top H(f(z)) J_f(z), \quad \forall z \in \mathbb{R}^m.$$

When the ambient manifold $\mathcal{N}$ is Euclidean, this reduces to:

$$G(z) = J_f(z)^\top J_f(z), \quad \forall z \in \mathbb{R}^m. \tag{2}$$

However, since we do not enforce any geometric constraints while pre-training the VAE, one can not expect that such a metric G is well-conditioned. Moreover, computationally, evaluating $G(z)$ for a given z requires one forward and one backward pass, which is prohibitively expensive.

Hence, we propose to directly "learn" a metric network in the family of metrics over the latent space that is smooth and guaranteed to be SPD, i.e., obeying Equation (2). From the PSD assumption of G, we define such a *Metric Family* via the following parameterization:

$$G^{-1}(z) = \sum_{i=1}^{M} L_i L_i^\top e^{-\frac{\|z - c_i\|}{T^2}} + \lambda I \tag{3}$$

where L_i, c_i, T are a trainable low-rank Cholesky factor, kernel centers, and kernel bandwidth, respectively. $G^{-1}(z)$ blends low-rank factors with Gaussian RBF's.

Let us denote the triplet of trainable metric parameters $(L_i, c_i, T)_{i=1}^M$ as ψ, and the matrix inverse of $G^{-1}(z)$ as $G_\psi(z)$. We then have a training loss that uses a pull-back metric $G_{\text{pb}}(z) = J^\top(z)J(z)$ w.r.t. the Frobenius norm coupled with a curvature variance regularizer:

$$\mathcal{L}(\psi) = \mathbb{E}_{z \sim q_\phi}[\|G_\psi(z) - G_{\text{pb}}(z)\|_F^2] + \lambda_{\text{curv}} \operatorname{Var}_{z \sim q_\phi}[\kappa_{\text{sec}}(z)]$$

where the expectation is w.r.t. the aggregate posterior of the VAE. The purpose of the sectional curvature variance regularizer is to prevent $G_\psi(z)$ from varying too wildly. In practice, the computationally favorable stochastic trace estimator by Skilling [1989] and Hutchinson [1989] is used over a mini-batch to get an unbiased trace estimate.

Additionally, we include another term $\mathcal{R}_{\text{cond}} = \mathbb{E}[\|\log \operatorname{diag}(L(\psi))\|_2^2]$ to keep eigenvalues within moderate magnitude. In all, the objective function for the metric family is as follows:

$$\mathcal{L}_{\text{metric}}(\psi) = \mathcal{L}(\psi) + \mathcal{R}_{\text{cond}}. \tag{4}$$

4.3 Manifold-Aware Latent Diffusion

After learning the regularized isometric metric G_ψ, our generative objective is to learn a score network $s_\varphi(z,t)$ such that running the reverse SDE on $(\mathcal{M}, G_\psi)$ produces samples whose push-forward distribution matches the data distribution $p_{\text{data}}(x)$. Stochastic differential equations (SDE) on the latent variable z_t can be generalized from the usual Euclidean reverse-SDE to a Riemannian manifold with metric $G_\psi(z)$ following Huang et al. [2022].

On a Riemannian manifold $(\mathcal{M}, G_\psi)$ with metric $G_\psi(z) = L_\psi(z) L_\psi^\top(z)$, true Brownian motion solves

$$\mathrm{d}z_t = \sqrt{2} L_\psi^\top(z_t)^{-1}\, \mathrm{d}w_t,$$

where w_t has a covariance G_ψ^{-1}. Instead of sampling in this space directly, one introduces a new coordinate

$$y_t = L_\psi^\top(z_t)^{-1} z_t,$$

for which the SDE becomes

$$\mathrm{d}y_t = \sqrt{2}\, \mathrm{d}w_t, \quad z_t = L_\psi^\top(z_t)^{-1} y_t.$$

Hence, the forward diffusion iteratively solves the system of linear equations $L_\psi(z_{t-1})^\top z = y_t$ for z to reconstruct z_t. The score loss

$$\mathcal{J}(\varphi) = \mathbb{E}\left[\left\| s_\varphi - \sqrt{2} \sigma_t^{-1} (L_\psi^\top)^{-1} w \right\|_{G_\psi}^2 \right] \tag{5}$$

reduces to the MSE in y-space, and the time-reversed SDE on z_t is

$$\mathrm{d}z_t = \left[2\ \mathrm{div}\big(G_\psi^{-1}\big) - s_\varphi(z_t, t) \right] \mathrm{d}t + \sqrt{2}\, L_\psi^\top(z_t)^{-1} \mathrm{d}\bar{w}_t. \tag{6}$$

4.4 Geodesic Computation

With the trained G in hand, we approximate the shortest paths (*geodesics*) between two latent codes $z_0, z_1 \in \mathbb{R}^m$. For a smooth curve $\gamma \colon [0,1] \to \mathcal{M}$, we define its *energy* via the functional

$$E[\gamma] \coloneqq \frac{1}{2} \int_0^1 \dot{\gamma}(t)^\top G(\gamma(t)) \dot{\gamma}(t)\, \mathrm{d}t, \quad \dot{\gamma}(t) = \frac{\mathrm{d}\gamma}{\mathrm{d}t}.$$

Geodesics are the curves that minimise E subject to the boundary conditions $\gamma(0) = z_0,\ \gamma(1) = z_1$. If the trained G is close to the true metric, the decoded geodesic should *lie on* $\mathcal{N}$. We discretize the domain of γ into N segments: $0 = t_0 < t_1 < \cdots < t_N = 1$, each of length $\Delta t = \frac{1}{N}$. We then represent the path by a sequence of points $\gamma_{\text{discrete}} = (z_0, \ldots, z_N)$ where the boundary pair (z_0, z_N) is fixed, and the interior points $z_1, \ldots, z_{N-1}$ are optimized. Then the discrete energy functional is as follows:

$$E_N[\gamma_{\text{discrete}}] = \sum_{k=0}^{N-1} \frac{1}{2} (z_{k+1} - z_k)^\top G\Big(\frac{z_{k+1} + z_k}{2}\Big) (z_{k+1} - z_k).$$

Gradient descent on the interior points drives E_N down until convergence to approximate geodesics. In practice, one can perform random-restart fits of the interior-point path, pick the best, and then refine the best candidate with a smaller learning rate and more iterations.

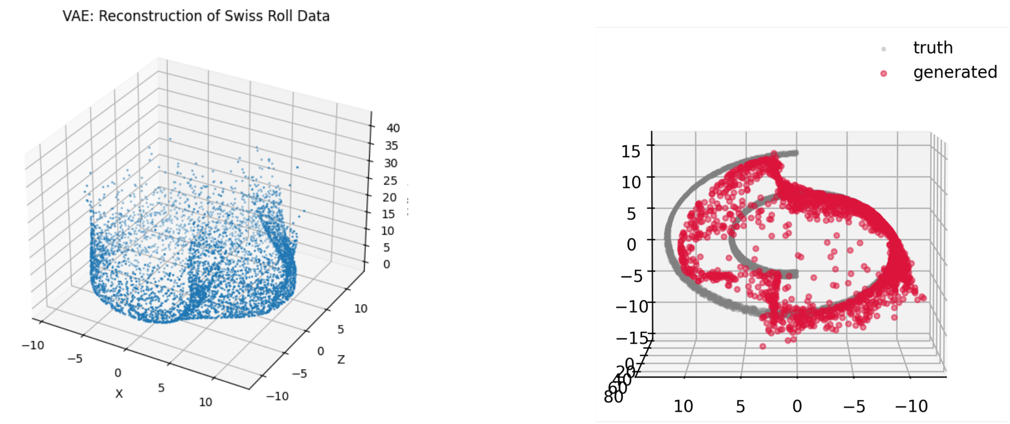
5 Results

By construction, the proposed metric network learns the latent Riemannian metric that preserves the geometry via isometry promotion. In this section, we first present numerical results that validate the method by qualitatively analyzing the decoded geodesic and the generated samples in the Swiss-roll manifold. Then, we further validate the model with the MNIST dataset. Note that we compare a standard VAE against our proposed framework (which uses that VAE as its pretrained backbone).

5.1 Swiss-roll Manifold

For the proof of concept, we generate a synthetic dataset that lies on the Swiss-roll manifold. After training the score network and the SPD metric network, we use them to generate samples and approximate the geodesics, respectively.

Generated Samples. Figure 1 shows the result of sample generation. It is clear that the generation from VAE completely ignores the manifold structure of the data. On the other hand, the samples from the proposed model are generally on the Swiss-roll manifold, which validates our proposed method.



(a) VAE with isotropic Gaussian noises. (b) Score-based diffusion with SDE.

Figure 1: Comparison of the generated samples between VAE and our proposed method.

Geodesic. Figure 2 shows the decoded geodesic on the latent manifold. The decoded geodesics are close to the data manifold. This implies that $G_\psi \approx G_{\text{pb}}$, which again validates the proposed model's capability in learning the latent Riemannian metric.

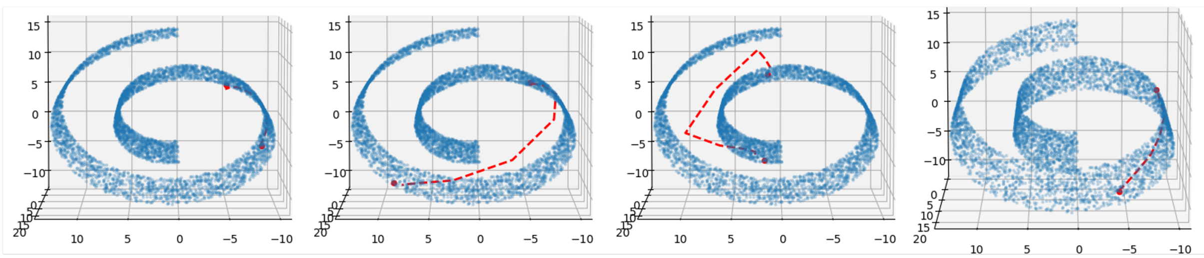


Figure 2: Visualization of Swiss-roll (scatter) with decoded geodesic (red line)

5.2 MNIST

Figure 3 presents the generated samples from the VAE and the proposed method that uses that VAE as its pretrained encoder-decoder. The reverse SDE was numerically solved with 500 iterations with the time step size being 0.001. Both models were able to generate high-quality samples. Then we test whether the model can generate samples that reflect the learned geometry by decoding points

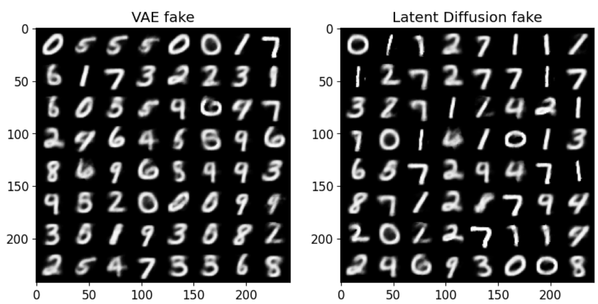


Figure 3: Generated samples from VAE (Left) and the proposed model (Right).

along the geodesic in latent space. Because the geodesic is the shortest path under the Riemannian metric, we expect two things: (1) the decoded geodesic samples will differ meaningfully from those obtained by simple linear interpolation, and (2) the sequence of generated samples will vary in a more coherent, interpretable way. The latent dimension for both VAE and the proposed method is set to 2 to visualize their latent representation. Figure 4 presents the latent space visualization of both models and the synthetic samples generated along linear and geodesic interpolations in the latent space. For the VAE, the decoded samples continuously deformed into a sharp-edged "1". However, in the second row of the left panel images, we can observe that "0" becomes the "average" of two digits but not an intermediate digit. On the other hand, the proposed method actually "visits" the intermediate digit 7, reflecting the fact that we are decoding a geodesic from 0 to the sharp-edged "1".

One can observe that the geodesic sometimes jumps abruptly. This likely reflects the inherent challenges of optimizing a nonlinear energy over a finite number of segments. Another explanation is that, in higher-dimensional latent manifolds, these jumps may smooth out naturally: digit clusters in 2D concentrate around $(0, 0)$ but additional dimensions will separate them more clearly, yielding a smoother geodesic.

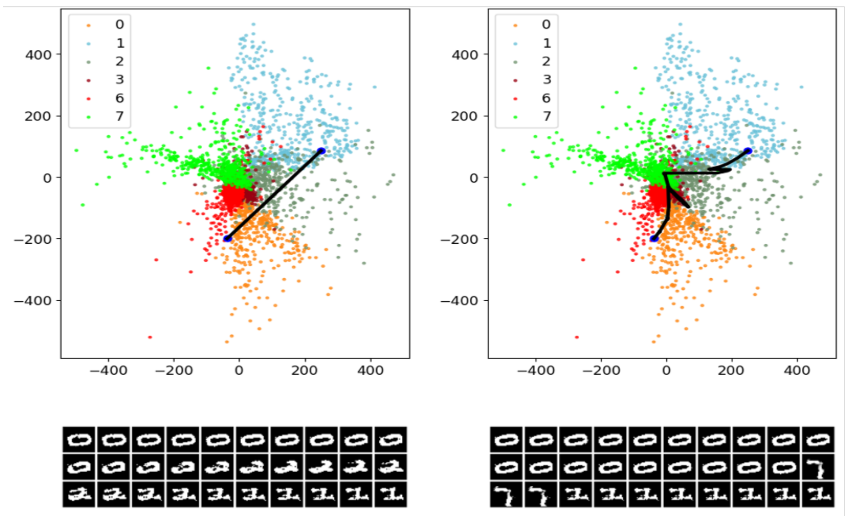


Figure 4: Visualization of the generated samples along linear interpolants in the latent space (Left) and the geodesic interpolants (Right). The figure shows the result of the latent space interpolations. The digits below are the decoded interpolated values from the left blue points to the right blue points. For clarity, we only plotted the latent representation of 6 different digits.

6 Conclusions

In this work, we proposed a novel manifold-aware generative framework that integrates learned SPD metrics with latent diffusion models. Our approach promotes local isometry by aligning the learned metric with the decoder's pullback, significantly reducing geometric distortion in the latent space. This design effectively captures complex data structures, resulting in more realistic sample generation and accurate geodesic estimation, as demonstrated on synthetic and real-world datasets.

Moving forward, this framework can be extended to handle more diverse and dynamically evolving data manifolds, offering potential applications in scientific computing, robotics, and other areas where data naturally exhibit non-trivial geometric structure. Our approach thus provides a promising direction for integrating geometric insights into deep generative modeling.

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